



AI-Powered Intelligent Triage Assistant

Universiti Teknologi Malaysia | Faculty of
Computing

SECJ 3553 - Artificial Intelligence

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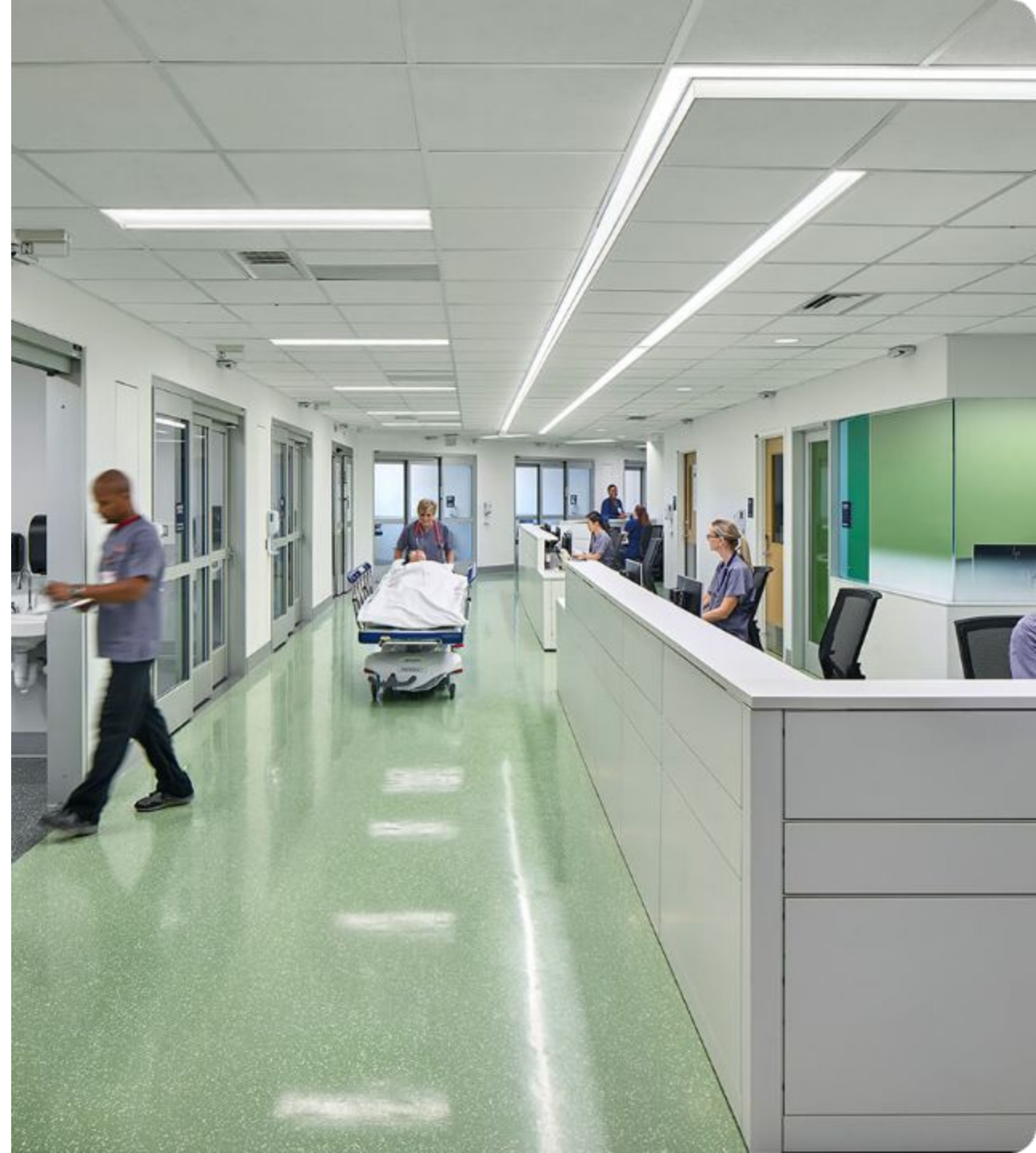
Project Overview

Streamlining Emergency Care with AI

AITA (AI-Powered Intelligent Triage Assistant) is designed to support the triage process in Emergency Departments (ED). By gathering patient data via sensors and manual entry, it analyzes severity levels to assist medical personnel.

Key Goals:

- Ensure critical cases receive immediate attention.
- Reduce patient waiting times.
- Automate bed allocation based on severity.





Knowledge Representation

Assignment 1: Logic Rules & Inference

Critical Severity Logic

Oxygen low	Body temp high	Heart rate high	Critical severity	(O ₂ ∧ Temp ∧ HR → Critical)
T	T	T	T	T
T	T	F	F	T
T	F	T	F	T
F	T	T	F	T
F	F	F	F	T

Heart rate high	Blood pressure low	Shock alert	(HR ∧ BP → Alert)
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

Condition deteriorate	Heart rate critical	Alert staff	(Cond ∨ HR → Alert)
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

Critical Prediction

The system uses First Order Logic to identify life-threatening conditions based on vital signs.

```
IF Oxygen < 90%
AND Temp > 38°C
AND HR > 110 bpm
THEN Severity = CRITICAL
```

Shock Alert

Immediate alerts are triggered for shock symptoms to notify the medical team instantly.

```
IF HR > 120 bpm
AND BP < 90/60 mmHg
THEN TRIGGER Shock_Alert
```

Critical Alert

If patient's condition is deteriorating OR heart rate becomes critical, then system will alert medical staff.

```
IF condition_deteriorate = TRUE OR
heart_rate_critical = TRUE, THEN alert_staff = TRUE
```

Resource & Maintenance Logic

Symptom match	Suggest diagnosis	Symptom -> Diagnosis
T	T	T
T	F	F
F	T	T
F	F	T

High Priority	Bed Availability	Assign Bed	Priority $\wedge$ BedAvail $\rightarrow$ Assign Bed
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

Usage Hours	High Error Rate	Maintenance Required	Hours $\vee$ Error Rate $\rightarrow$ Maintenance Required
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T



Diagnosis

Matches symptom patterns to known diseases to suggest preliminary diagnoses to staff.



Bed Allocation

IF Patient is High Priority **AND** Bed is Available **THEN** Assign Bed immediately.



Maintenance

Predicts equipment failure if usage hours exceed threshold or error rates are high.



State Space Search Model

Assignment 2: State Space Search Model

State Space Search Model

Initial State

Check Equipment → Verify Operational Status

Actions

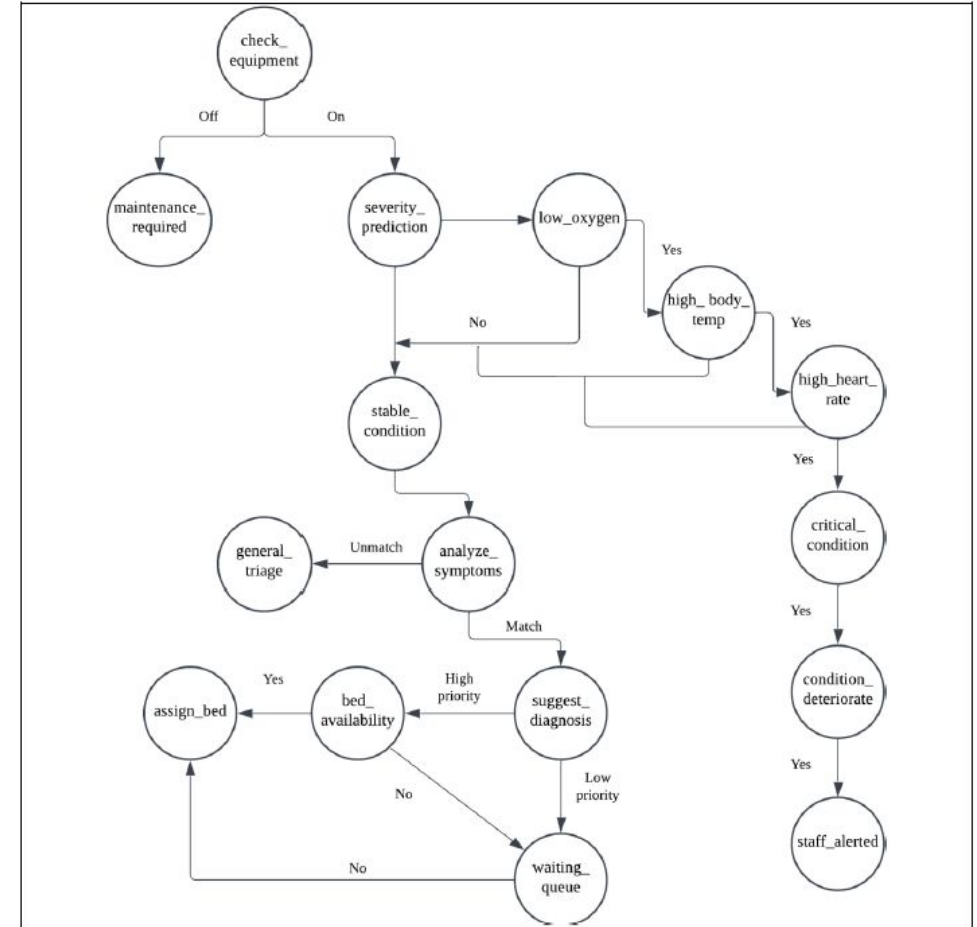
check_oxygen, check_body_temp, check_heart_rate,
analyze_symptoms, predict_severity, check_bed_availability,
trigger_staff_alert, request_maintenance

Goal State

Patient Assigned Bed **OR** Staff Alerted (if
critical/deteriorating)

Path Cost

Yes, No, On, Off, Match, Unmatch, High priority, Low priority





PEAS Model

ASSIGNMENT 3: INTELLIGENT AGENT

PEAS Model Representation

Performance & Environment

- **Performance:** Accuracy of severity scoring, reduction in wait times, correct prioritization.
- **Environment:** Emergency Department, Triage Station, Hospital Information System (HIS).

Actuators & Sensors

- **Actuators:** Triage Dashboard, Visual/Audio Alerts, Database Updates.
- **Sensors:** Thermometer, Pulse Oximeter, BP Monitor, Keyboard Entry.

Diagram of Representation

1. Sensors to Inference

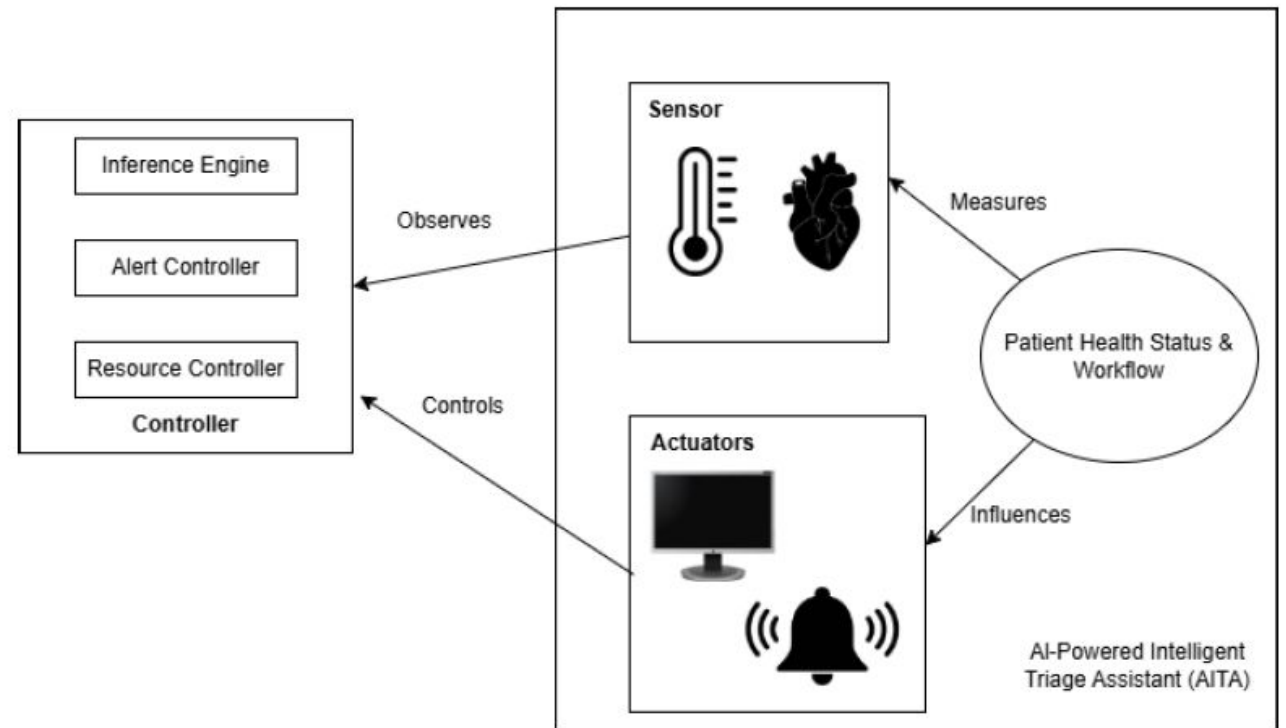
Sensors measure patient health and workflow data, feeding it into the AI Inference Engine.

2. Intelligent Processing

The AITA (AI Assistant) observes the data and determines the state (Critical vs Stable).

3. Controllers & Actuators

The Inference Engine influences Alert and Resource Controllers, which trigger Actuators (Dashboard, Alerts) to affect the environment.





PROOF OF CONCEPT

FINAL ASSIGNMENT : POC

FIGMA PROTOTYPE

1. LOGIN PAGE

Secure Network: ED-MESH-01 | ED Bays Available: 9/9 | System Online 13:05

AITA
INTELLIGENCE

Welcome Back

Please verify your identity to access ED Triage

STAFF ID

PASSWORD

Biometric Login

Authorized personnel only — UTM Group 4

EXPLANATION

This login page is designed to be more than just a security gate; it is an informational dashboard. By displaying the number of available ED bays and system status in the header, we allow doctors to make quick capacity assessments immediately. The interface prioritizes biometric login to reduce friction for busy medical professionals while maintaining strict security protocols..

FIGMA PROTOTYPE

2. ED TRIAGE QUEUE

The interface is titled "ED Triage Queue" and includes a header with system status: "AITA System | ED-MESH-01 | ED Bays Available: 9/9 | System Online 15:51". Below the header, there is a summary bar showing "Waiting: 12 • 2 Critical • 4 Emergent". The main content area displays a list of patients in a queue, each with a numbered card, patient name, level, symptoms, wait time, and an action button.

Queue Number	Patient Name	Level	Symptoms	Wait Time	Action
1	Li Qiang	Level 1: Critical	Chest pain > 20 min • Hx of MI • Diaphoresis	00:03	Triage
2	Arthur Dent	Level 1: Critical	Anaphylaxis • Airway compromise • SpO2 88%	00:08	Triage
3	Wang Fang	Level 2: Emergent	Severe abdominal pain (RLQ) • Fever 38.5°C	00:25	Assign
4	Sarah Jenkins	Level 2: Emergent	Sudden onset worst headache • BP 190/110	00:42	Assign
9	Robert Taylor	Level 2: Emergent	Syncopal • Head injury on fall	01:20	Assign
11	Thomas Moore	Level 2: Emergent	Palpitations • Lightheadedness	00:50	Assign

EXPLANATION

This triage queue is designed for emergency department(ED) medical staff to manage and prioritise patients awaiting treatment by providing real-time view of the triage queue. It enable medical staff to quickly identify critical and emergent cases, review key clinical information, track waiting times and take immediate actions such as triaging high-risk patients or assigning patients to available ED resources.

FIGMA PROTOTYPE

3. REGISTER NEW PATIENT

 Register New Patient

PATIENT ID

#013 (Auto-generated)

FULL NAME *

Enter patient name

AGE *

Age

GENDER

Male

WEIGHT (KG) *

e.g., 70

KNOWN ALLERGIES

None

CHIEF COMPLAINT *

Describe presenting symptoms and history...

INITIAL VITALS *

HR

bpm

120/80

mmHg

98

%

36.5

°C

 Register New Patient

CHIEF COMPLAINT *

Describe presenting symptoms and history...

INITIAL VITALS *

HR

bpm

120/80

mmHg

98

%

36.5

°C

 AI Pre-Triage Analysis


Waiting for input...
Enter chief complaint and vitals for AI analysis

Cancel

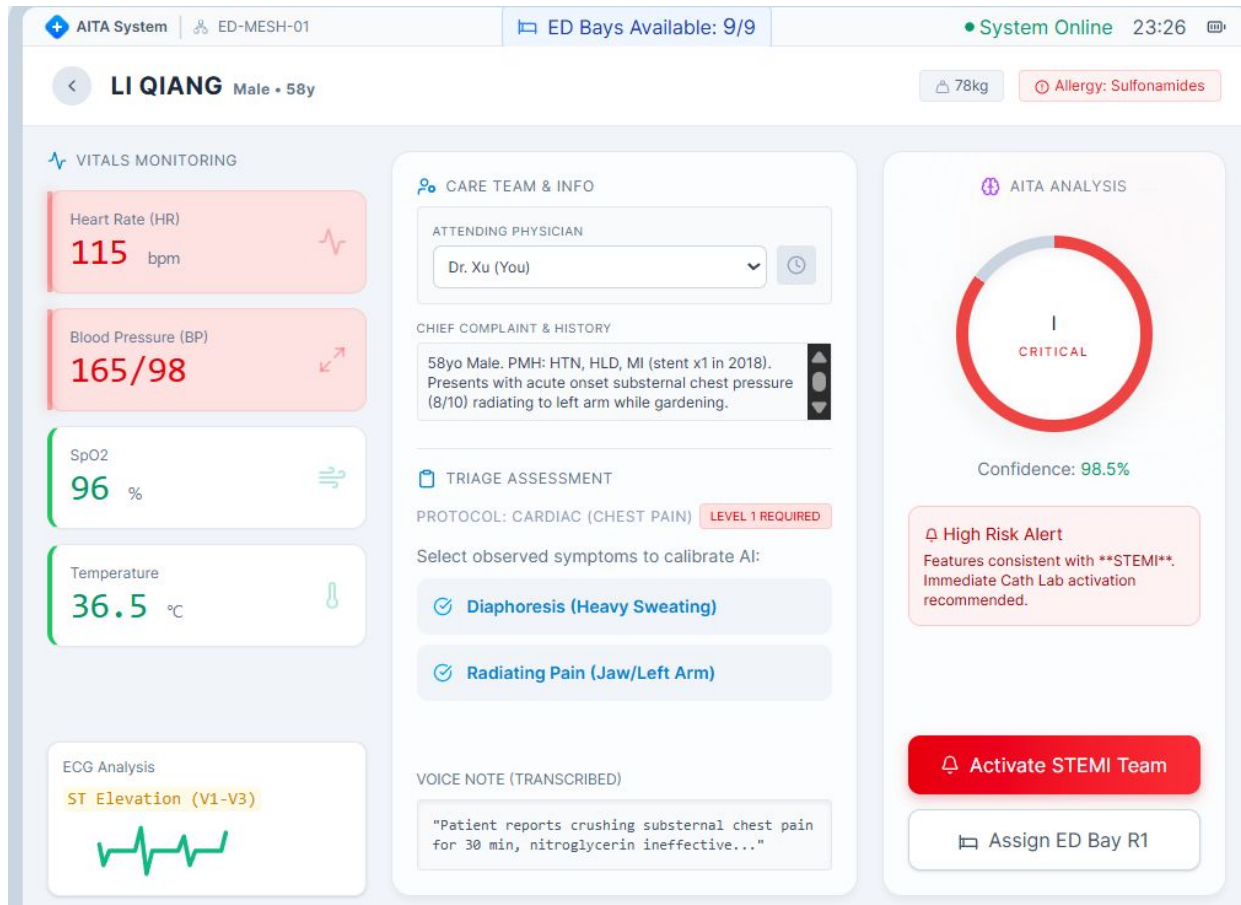
Analyze & Add

EXPLANATION

This registration interface is designed to streamline the patient intake process by capturing essential demographics alongside critical clinical indicators like chief complaints and initial vitals. It features an integrated "AI Pre-Triage Analysis" module that processes this data in real-time, enabling the system to assess severity levels immediately during data entry. This ensures that potential high-risk cases are identified the moment they are registered, allowing for faster transition to the triage queue and more efficient resource allocation.

FIGMA PROTOTYPE

4. VITALS



EXPLANATION

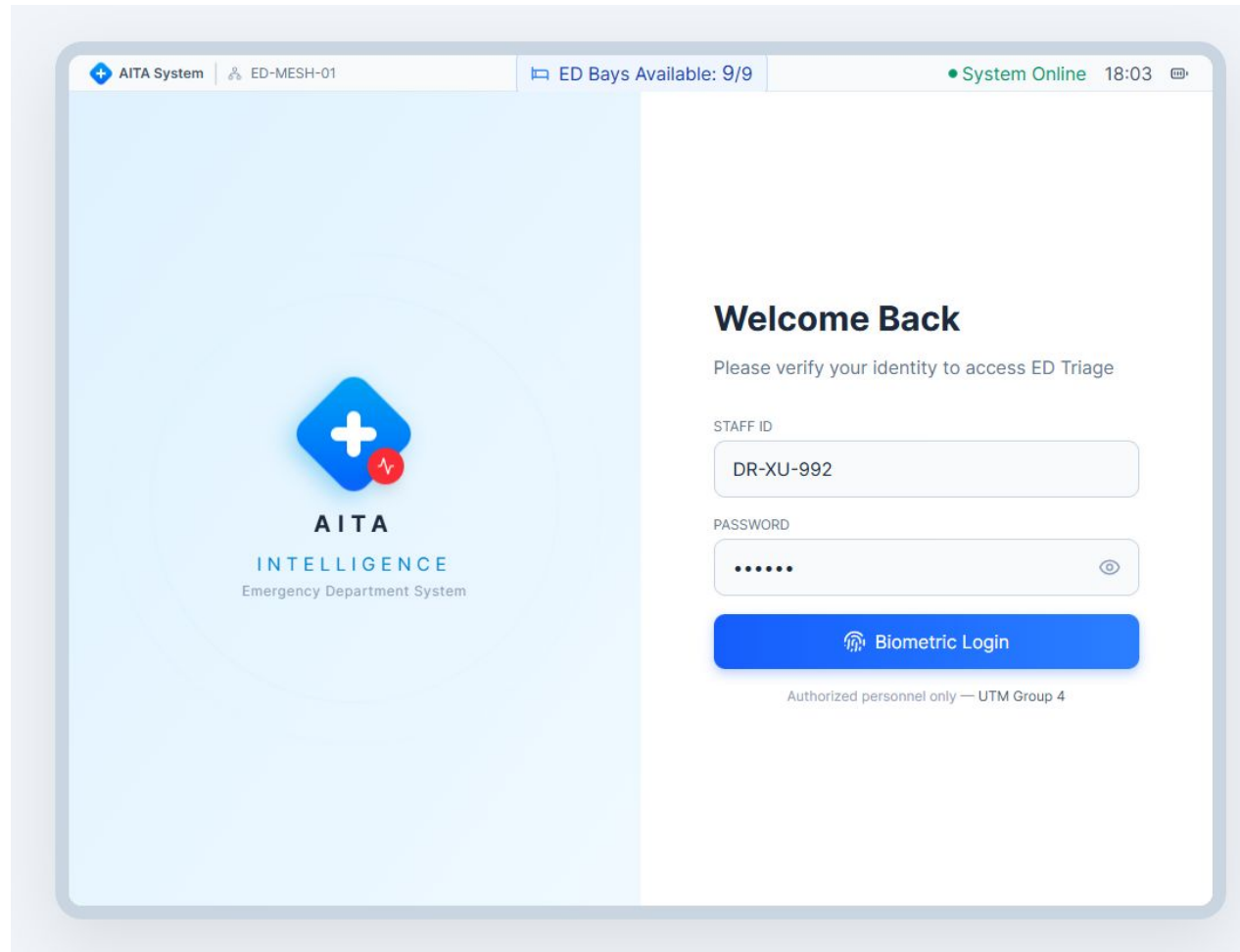
The Vitals Monitoring page serves as a real-time dashboard for medical staff to monitor and respond to emergency patient data. The page displays live health metrics like heart rate, blood pressure, and oxygen levels, while the AITA uses these data to instantly determine if a patient is in a critical state. Beyond just showing numbers, the system helps with decision-making by identifying dangerous symptoms, issuing high-risk alerts for conditions like heart attacks, and allowing staff to quickly perform essential actions like activating emergency teams or assigning a hospital bed.

Team Contribution

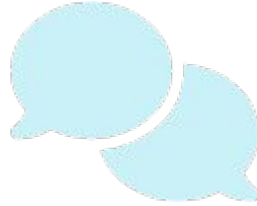
Peer review on Team Working (TW) - All members actively participated.

Name	Matric ID	Role	Score
Elijah She Yu Sheng	A23CS0073	Member	4/4 (Active)
Lau Yee Wen	A23CS0099	Member	4/4 (Active)
Cheryl Cheong Kah Voon	A23CS0060	Member	4/4 (Active)
Muhammad Adam Bin Razali	A23CS0116	Member	4/4 (Active)

FIGMA PROTOTYPE



FIGMA VIEWING LINK: [ED Triage Application](#)



Thank You

Any Questions?



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